

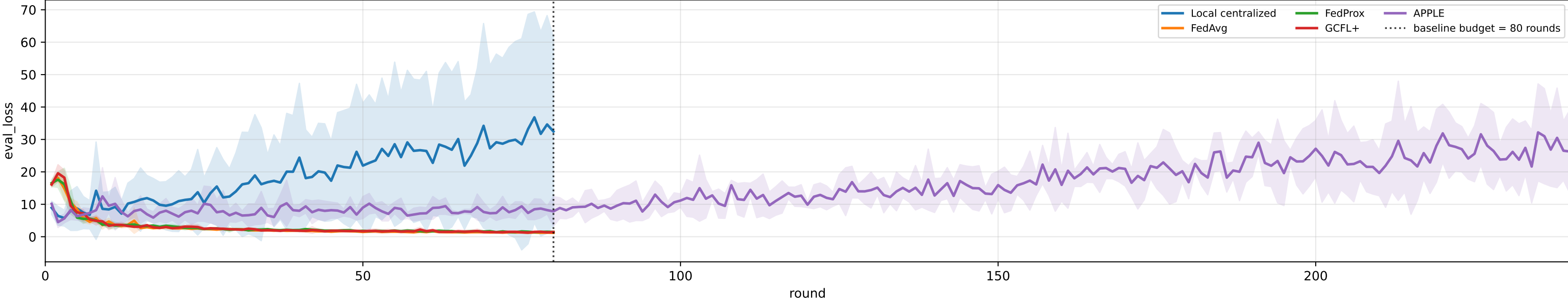
Controlled q-label heterogeneity: High | aggregated / pooled performance over 5 clients |
model=mcwauto_layers6_lr0.001_wd0.0001_d0.1_hd64_egoTrue_bs64

family=q_task_cycle2 | n_clients=1 | cycle2=visible 4.6%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
family=q_task_cycle3 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 5.0%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
family=q_task_cycle4 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 5.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
family=q_task_cycle5 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 5.1%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
family=q_task_cycle6 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 5.0%±0.0% / true 6.3%
heterogeneity=high | gamma=0.80 | task_jsd_mean=0.399 | task_jsd_median=0.399 | task_jsd_max=0.403 | family_centroid_jsd_mean=0.399

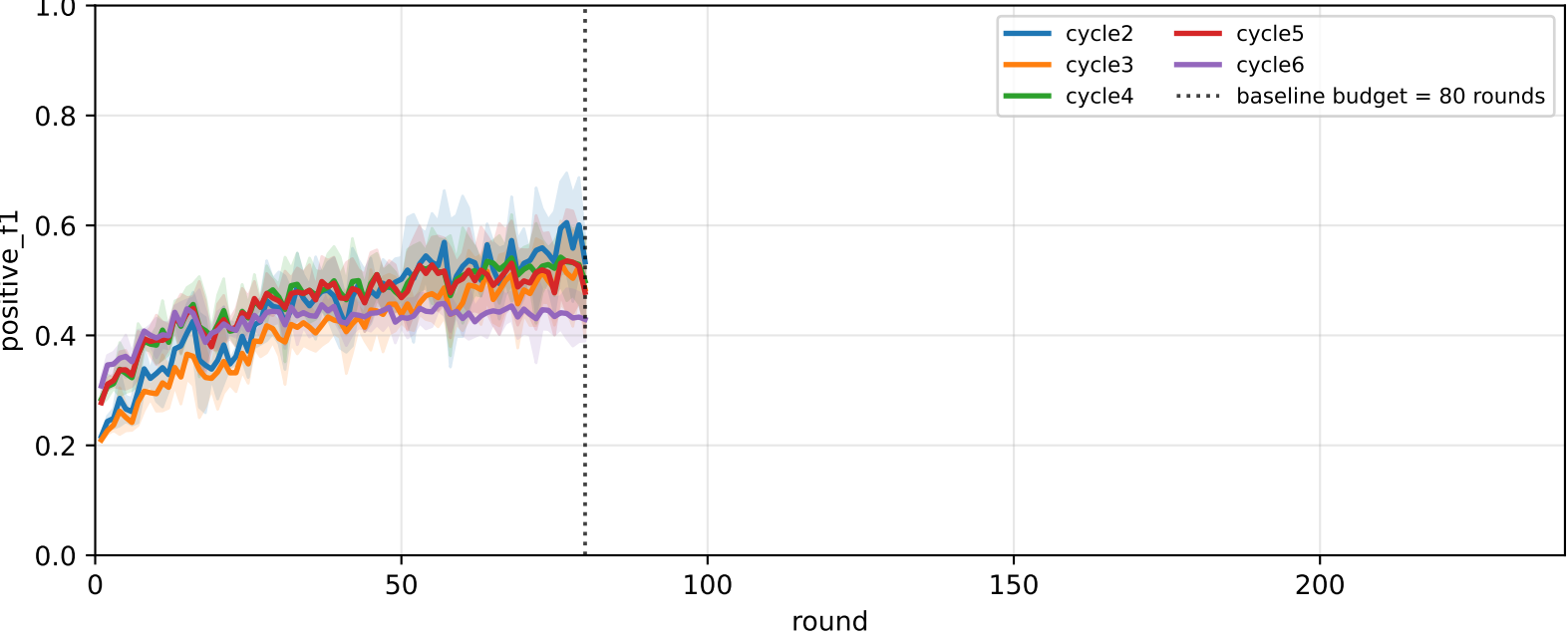
Best logged test performance (all methods 80 rounds; APPLE dr_lr=0.001, mu=0.01, Lfrac=0.3)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	29.1% ± 5.1%	29.5% ± 5.3%	22.1% ± 2.8%	26.7% ± 3.4%	26.9% ± 3.1%	26.5% ± 3.3%	30.2% ± 3.0%
FedAvg	54.6% ± 5.2%	55.0% ± 5.3%	50.2% ± 5.4%	60.6% ± 7.5%	56.9% ± 5.9%	57.5% ± 6.2%	49.7% ± 4.8%
FedProx	52.8% ± 2.9%	53.2% ± 2.8%	53.5% ± 2.9%	55.1% ± 3.6%	54.7% ± 5.7%	53.5% ± 6.0%	49.1% ± 1.1%
GCFL+	54.1% ± 1.1%	54.7% ± 1.3%	53.8% ± 4.5%	58.5% ± 4.0%	54.0% ± 4.5%	57.3% ± 5.5%	49.9% ± 0.9%
APPLE	31.3% ± 9.1%	31.3% ± 9.0%	26.4% ± 8.8%	31.7% ± 9.6%	32.3% ± 9.9%	31.3% ± 9.2%	34.4% ± 8.0%

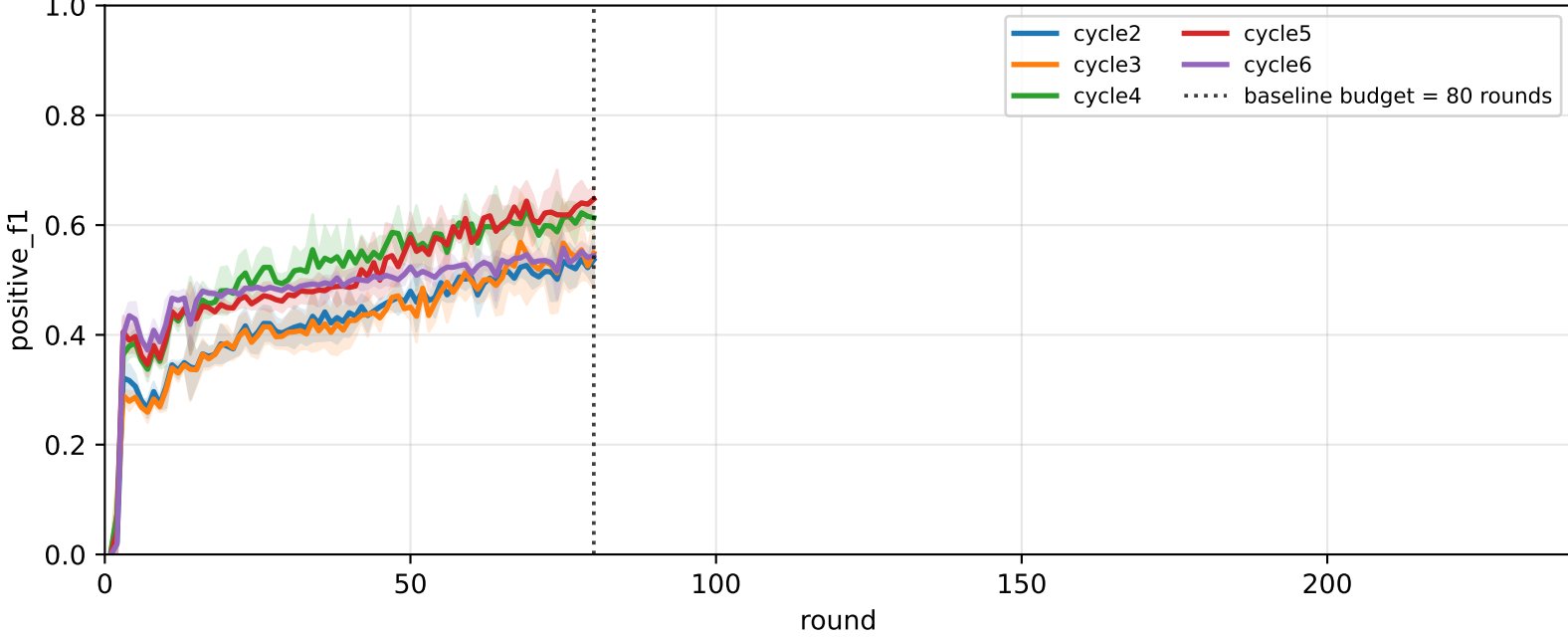
Validation loss comparison (all methods 80 rounds; APPLE dr_lr=0.001, mu=0.01, Lfrac=0.3)



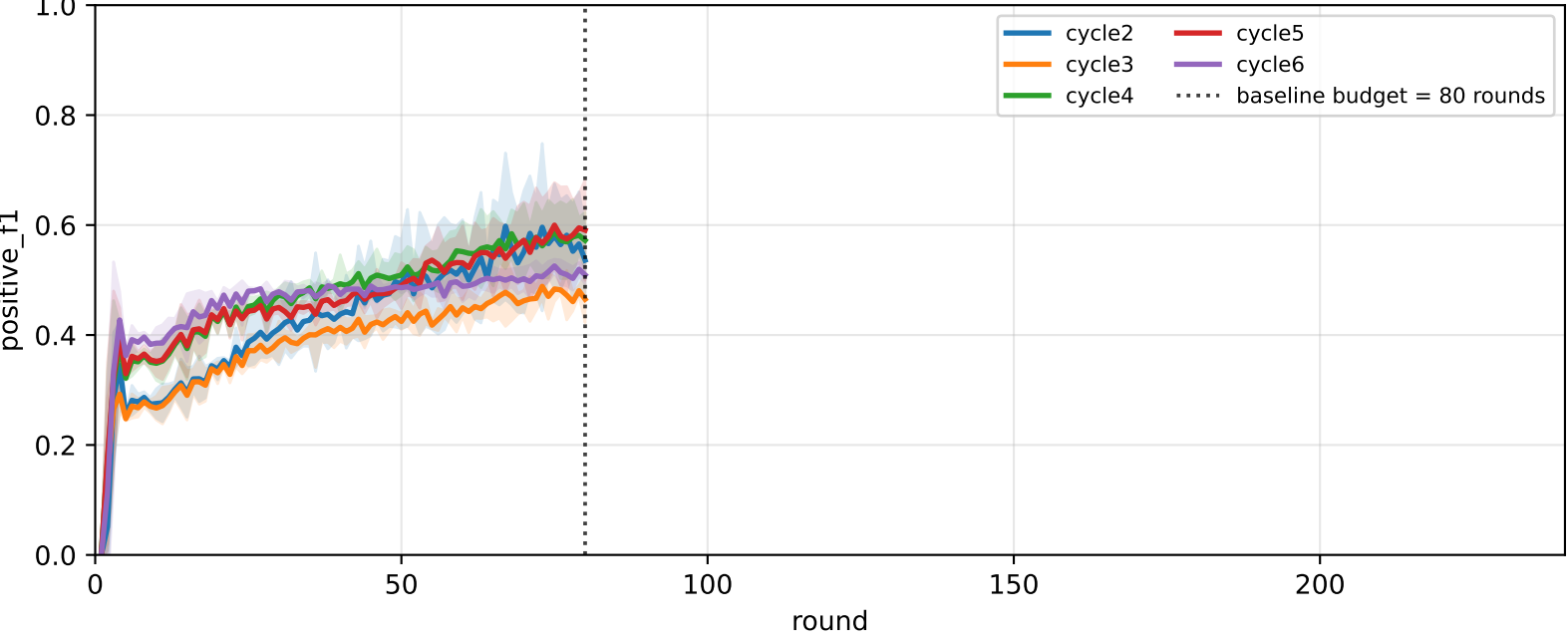
Pooled local centralized positive F1



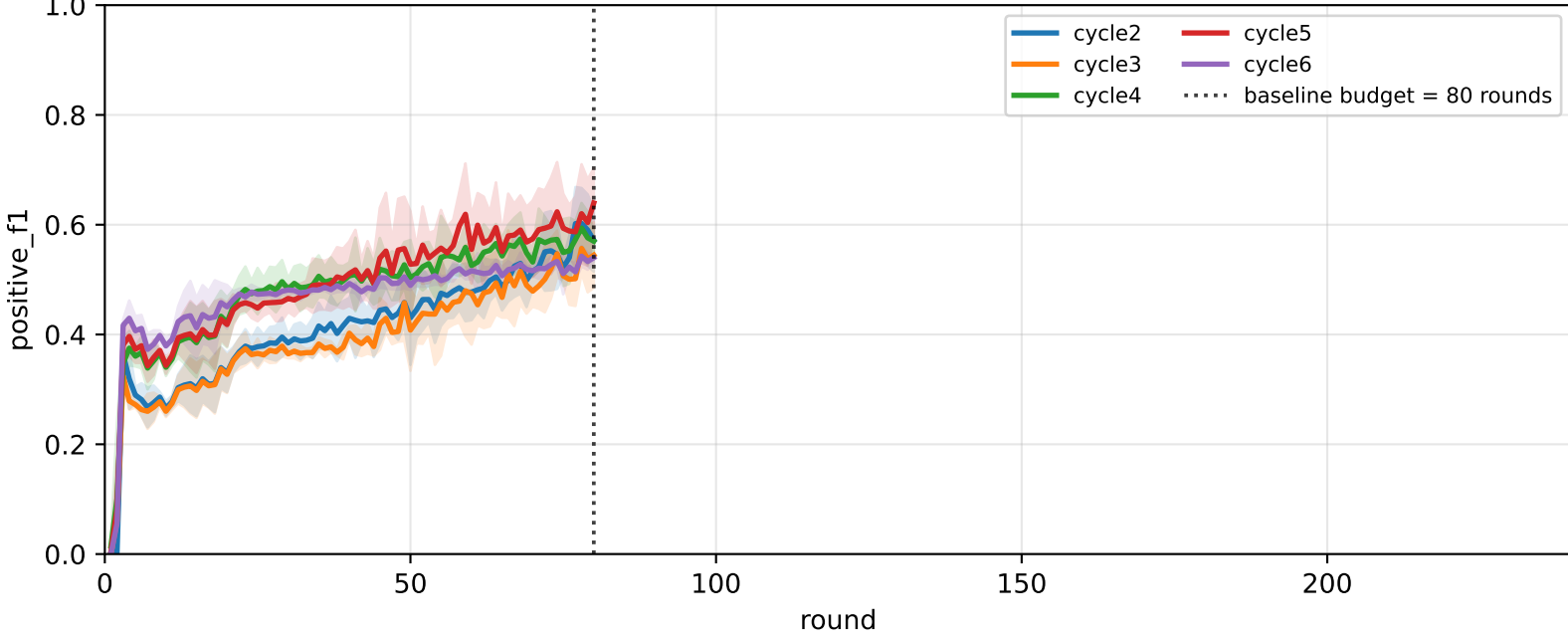
Pooled FedAvg positive F1



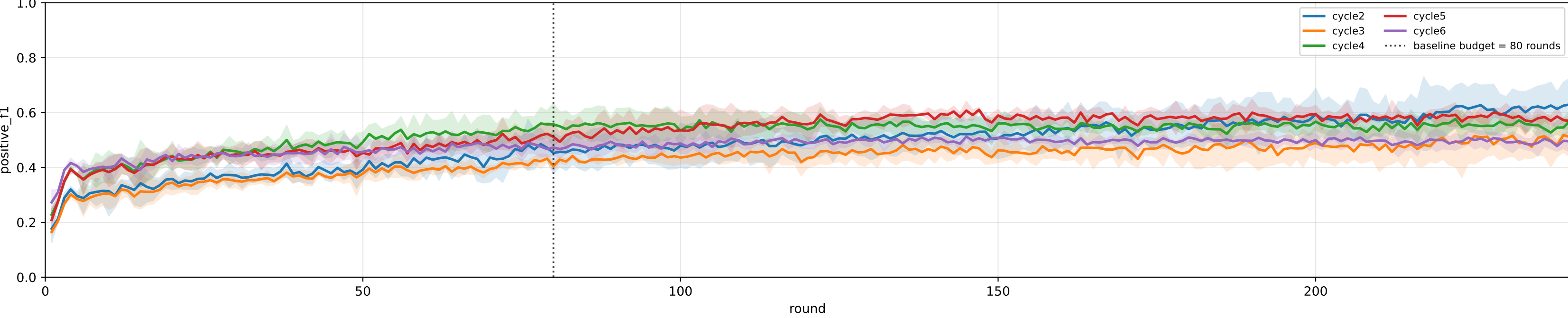
Pooled FedProx positive F1



Pooled GCFL+ positive F1



Pooled APPLE positive F1 (after-local selected model, equal 240-round budget)



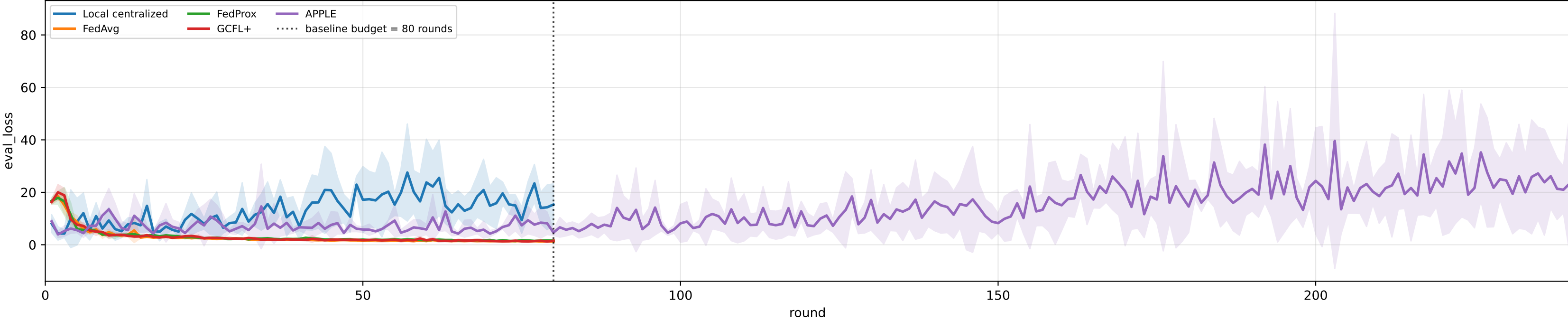
Controlled q-label heterogeneity: High | family aggregated / pooled performance |
family=q_task_cycle2 | n_clients=1 |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=q_task_cycle2 | n_clients=1 | cycle2=visible 4.6%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% |
cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3% |
heterogeneity=high | gamma=0.80 | task_jsd_mean=0.399 | task_jsd_median=0.399 | task_jsd_max=0.403 | family_centroid_jsd_mean=0.399

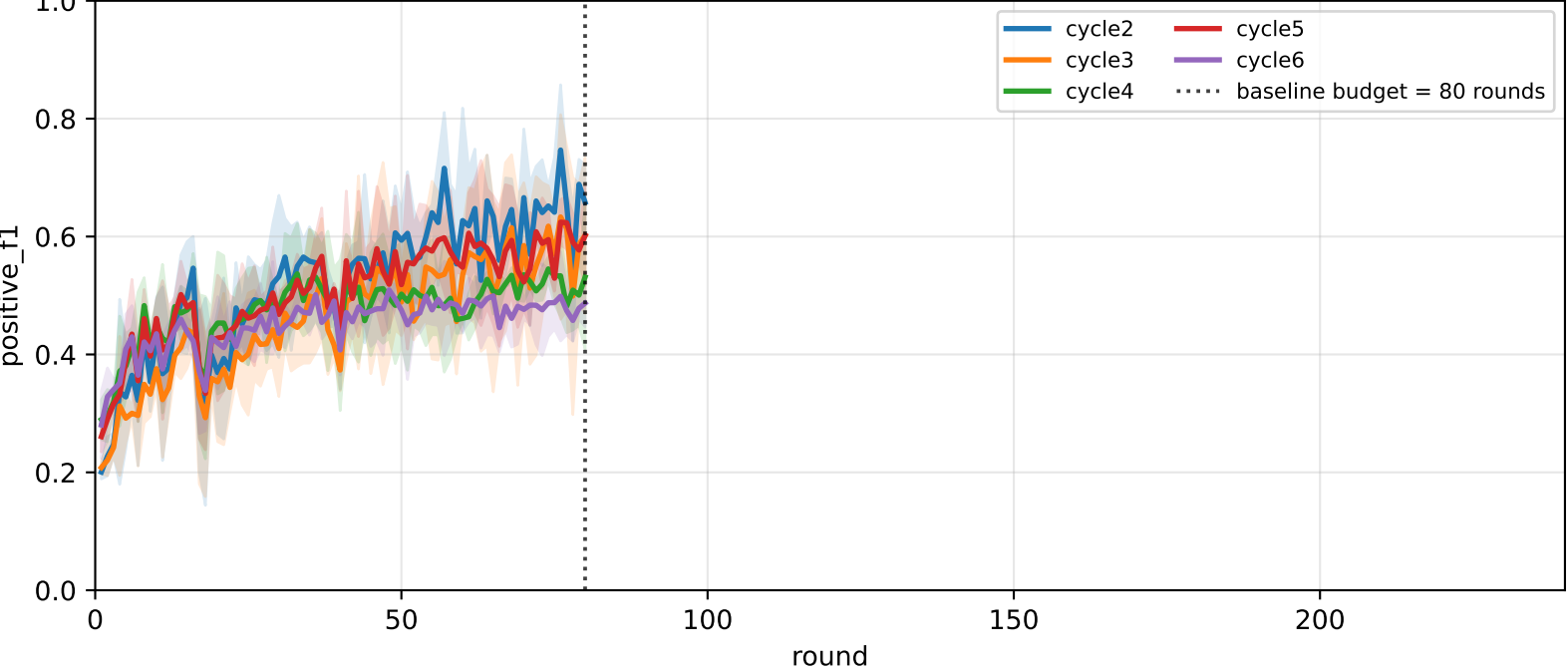
Best logged test performance (all methods 80 rounds; APPLE dr_lr=0.001, mu=0.01, Lfrac=0.3)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	32.7% ± 12.1%	33.2% ± 12.8%	31.2% ± 16.3%	35.6% ± 15.5%	35.8% ± 15.4%	31.2% ± 10.3%	32.3% ± 6.7%
FedAvg	54.5% ± 5.3%	55.0% ± 5.4%	50.1% ± 5.9%	60.4% ± 7.5%	57.0% ± 6.0%	57.5% ± 6.1%	49.7% ± 4.8%
FedProx	52.9% ± 2.8%	53.3% ± 2.7%	53.6% ± 3.3%	55.0% ± 3.6%	54.9% ± 5.7%	53.7% ± 6.0%	49.4% ± 1.1%
GCFL+	54.6% ± 1.2%	55.1% ± 1.4%	54.0% ± 4.1%	59.1% ± 4.2%	54.4% ± 4.2%	57.9% ± 5.4%	50.3% ± 0.8%
APPLE	32.0% ± 8.9%	32.0% ± 8.8%	26.7% ± 8.6%	32.6% ± 9.1%	32.8% ± 9.7%	31.8% ± 8.8%	35.9% ± 8.0%

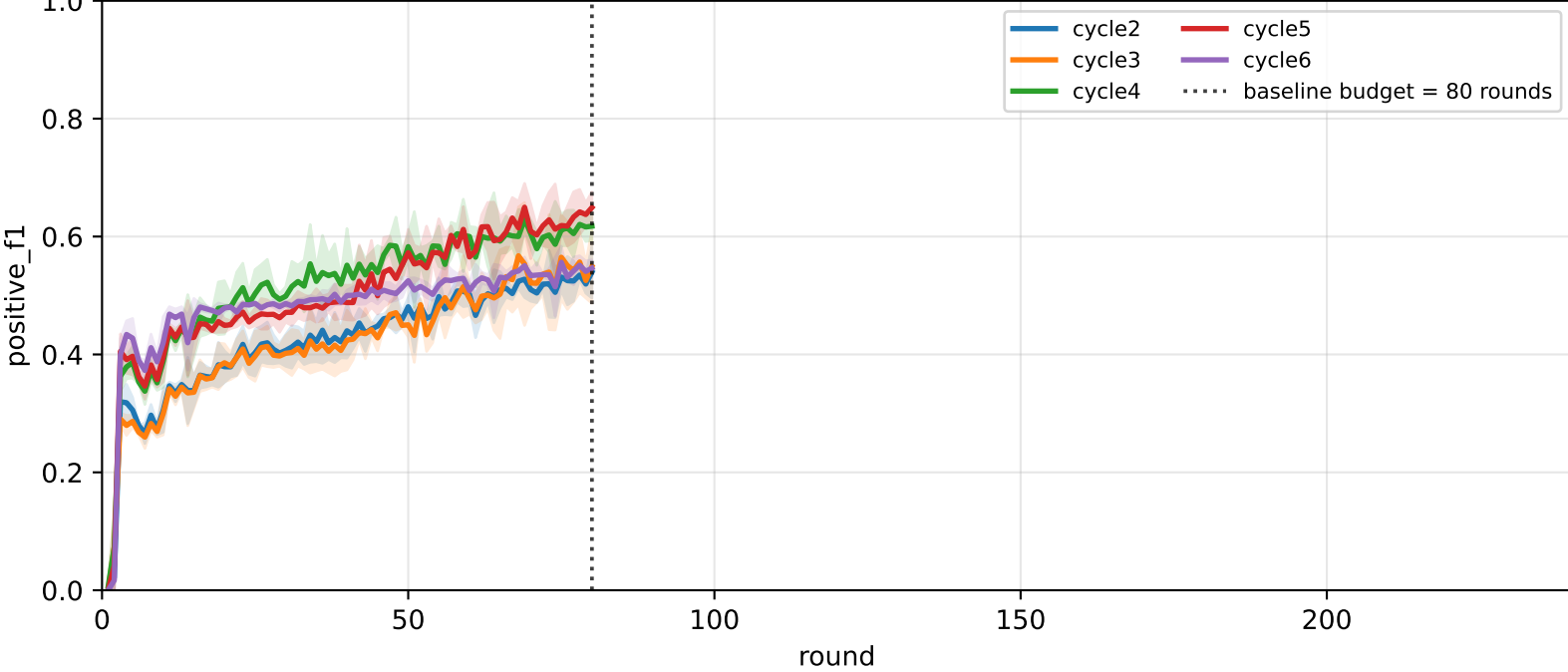
Validation loss comparison (all methods 80 rounds; APPLE dr_lr=0.001, mu=0.01, Lfrac=0.3)



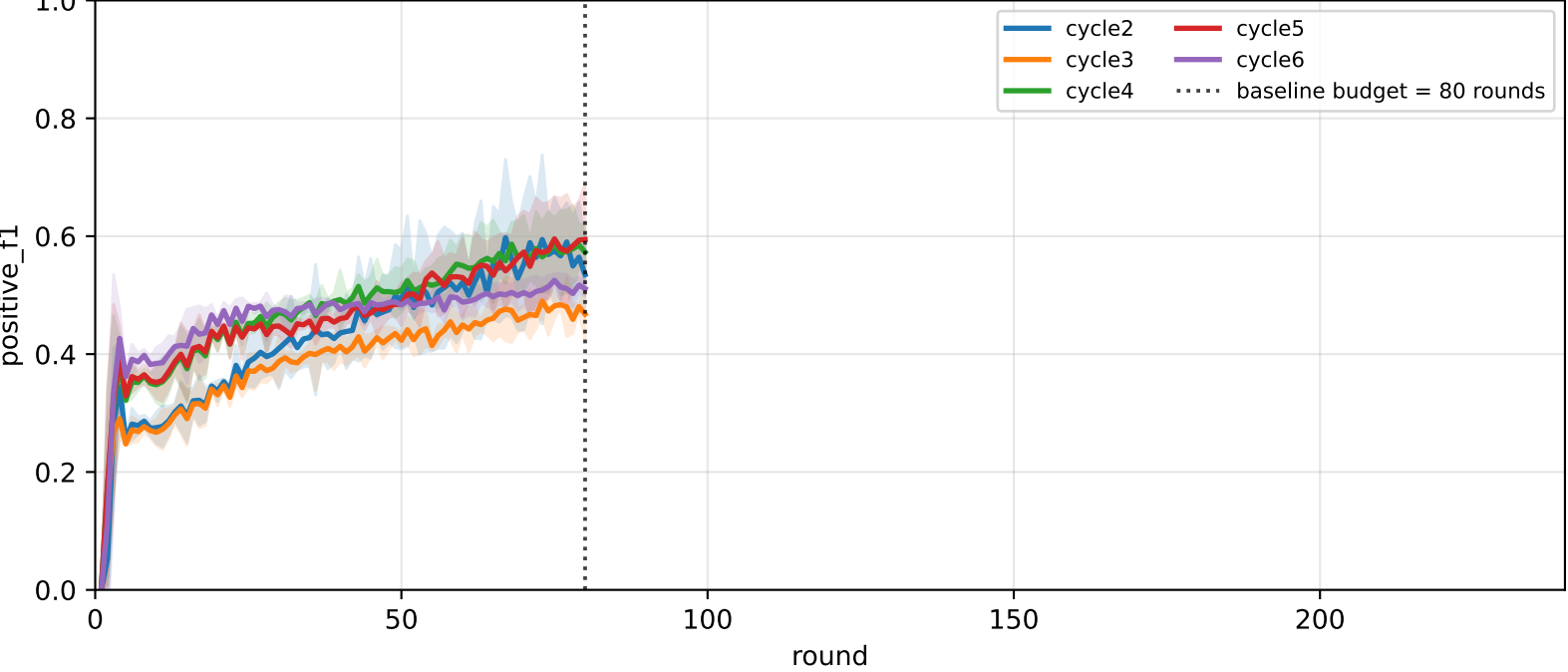
Pooled local centralized positive F1



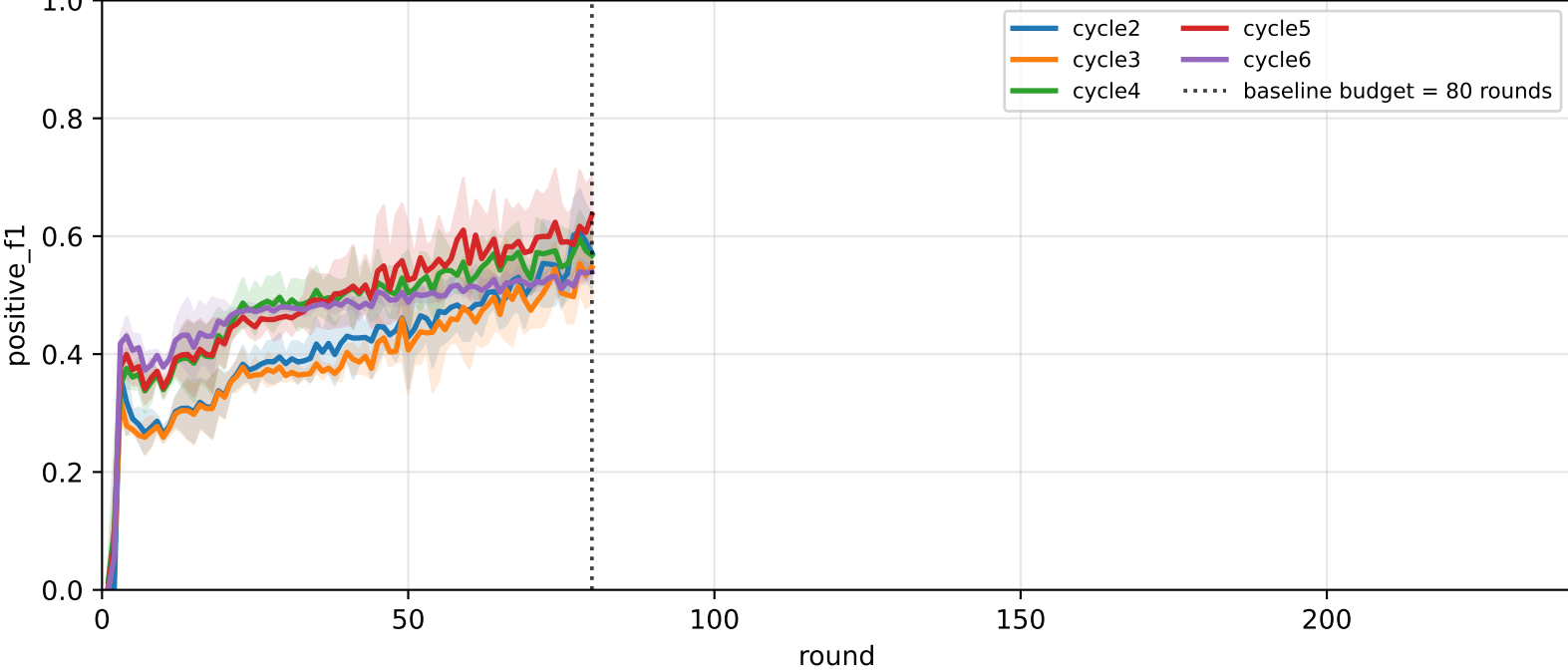
Pooled FedAvg positive F1



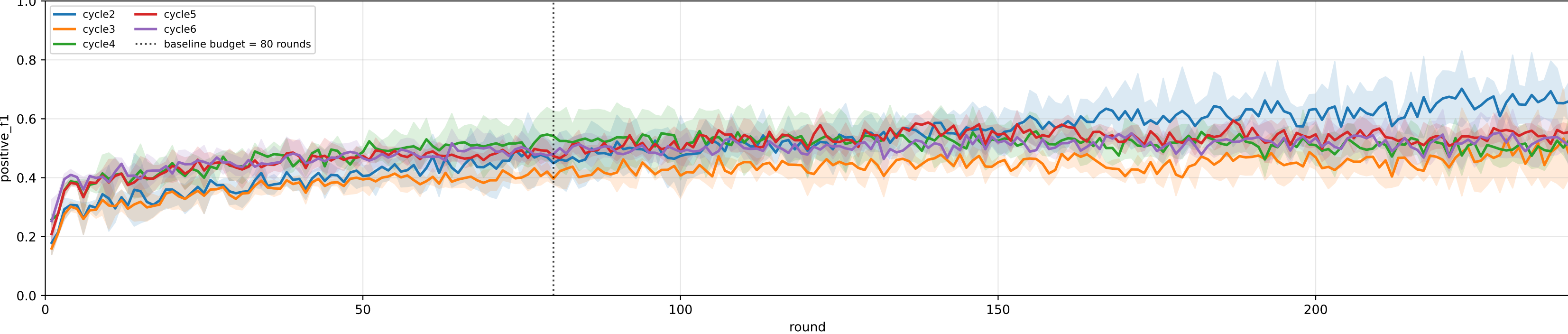
Pooled FedProx positive F1



Pooled GCFL+ positive F1



Pooled APPLE positive F1 (after-local selected model, equal 240-round budget)



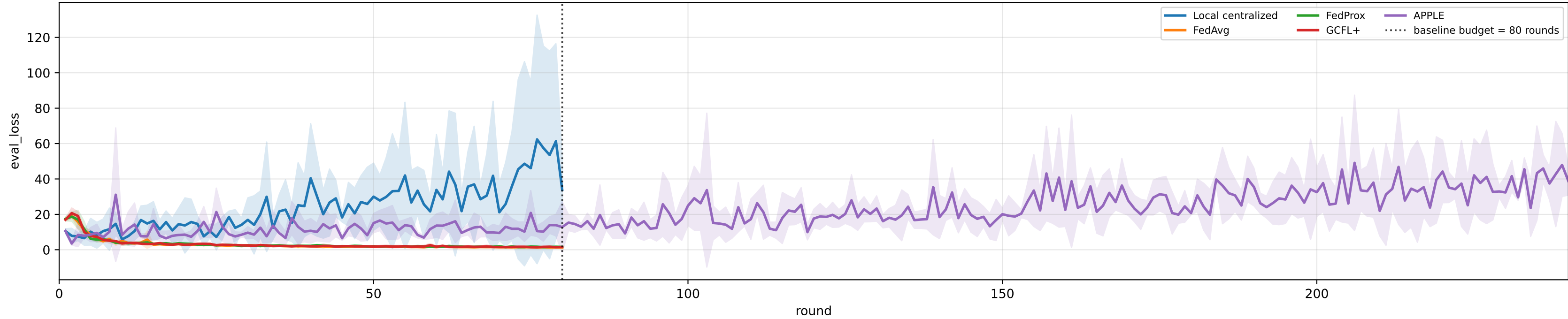
Controlled q-label heterogeneity: High | family aggregated / pooled performance |
family=q_task_cycle3 | n_clients=1 |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=q_task_cycle3 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 5.0%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% |
cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3% |
heterogeneity=high | gamma=0.80 | task_jsd_mean=0.399 | task_jsd_median=0.399 | task_jsd_max=0.403 | family_centroid_jsd_mean=0.399

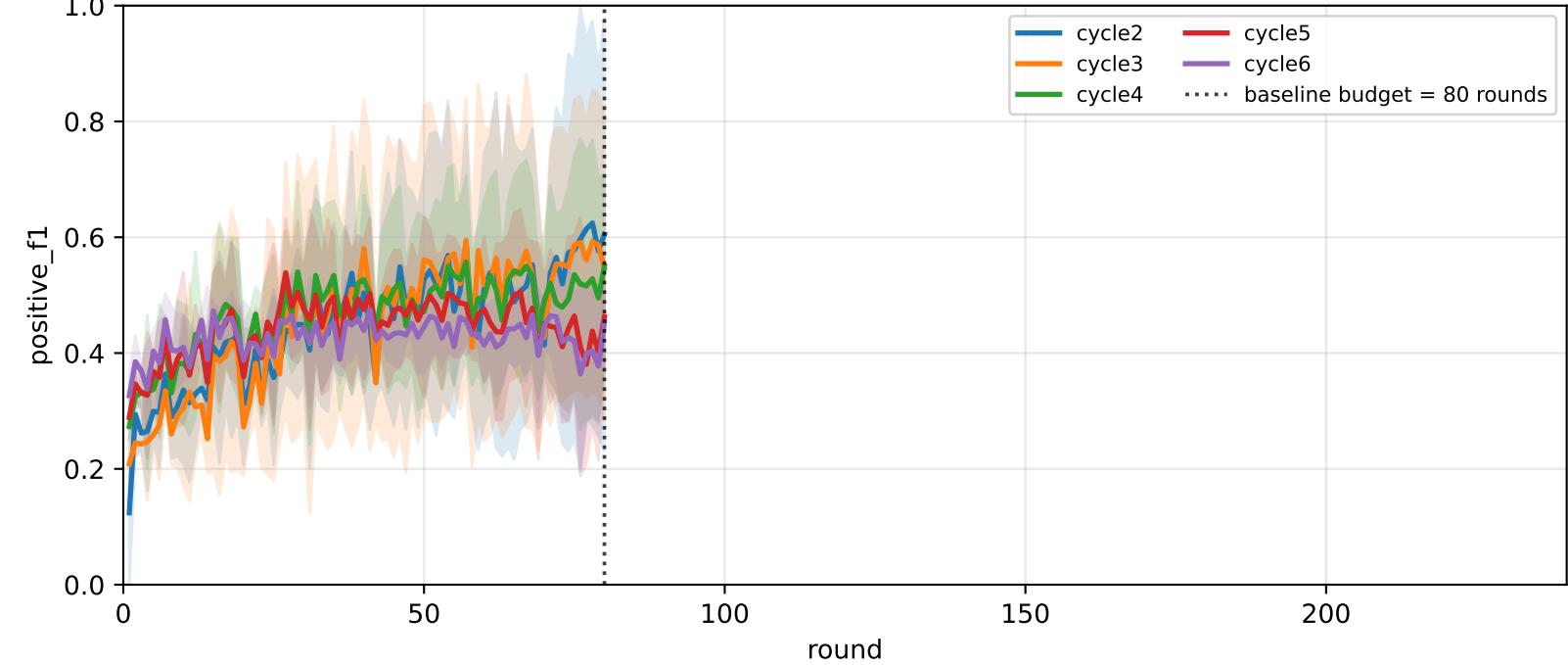
Best logged test performance (all methods 80 rounds; APPLE dr_lr=0.001, mu=0.01, Lfrac=0.3)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	24.6% ± 3.5%	24.6% ± 3.5%	20.2% ± 3.0%	23.5% ± 4.2%	24.7% ± 3.6%	25.1% ± 3.4%	29.3% ± 3.3%
FedAvg	54.6% ± 5.7%	55.1% ± 5.8%	50.3% ± 5.8%	60.9% ± 8.5%	56.9% ± 6.3%	57.5% ± 6.3%	49.9% ± 5.1%
FedProx	52.8% ± 2.6%	53.2% ± 2.5%	53.4% ± 3.2%	55.1% ± 3.4%	54.8% ± 5.3%	53.4% ± 5.6%	49.2% ± 0.9%
GCFL+	54.3% ± 1.3%	54.9% ± 1.4%	54.3% ± 4.4%	58.9% ± 4.0%	54.1% ± 5.0%	57.5% ± 5.9%	49.6% ± 0.7%
APPLE	31.4% ± 8.7%	31.3% ± 8.7%	26.8% ± 8.6%	31.5% ± 9.6%	32.5% ± 10.3%	32.2% ± 9.2%	33.7% ± 6.0%

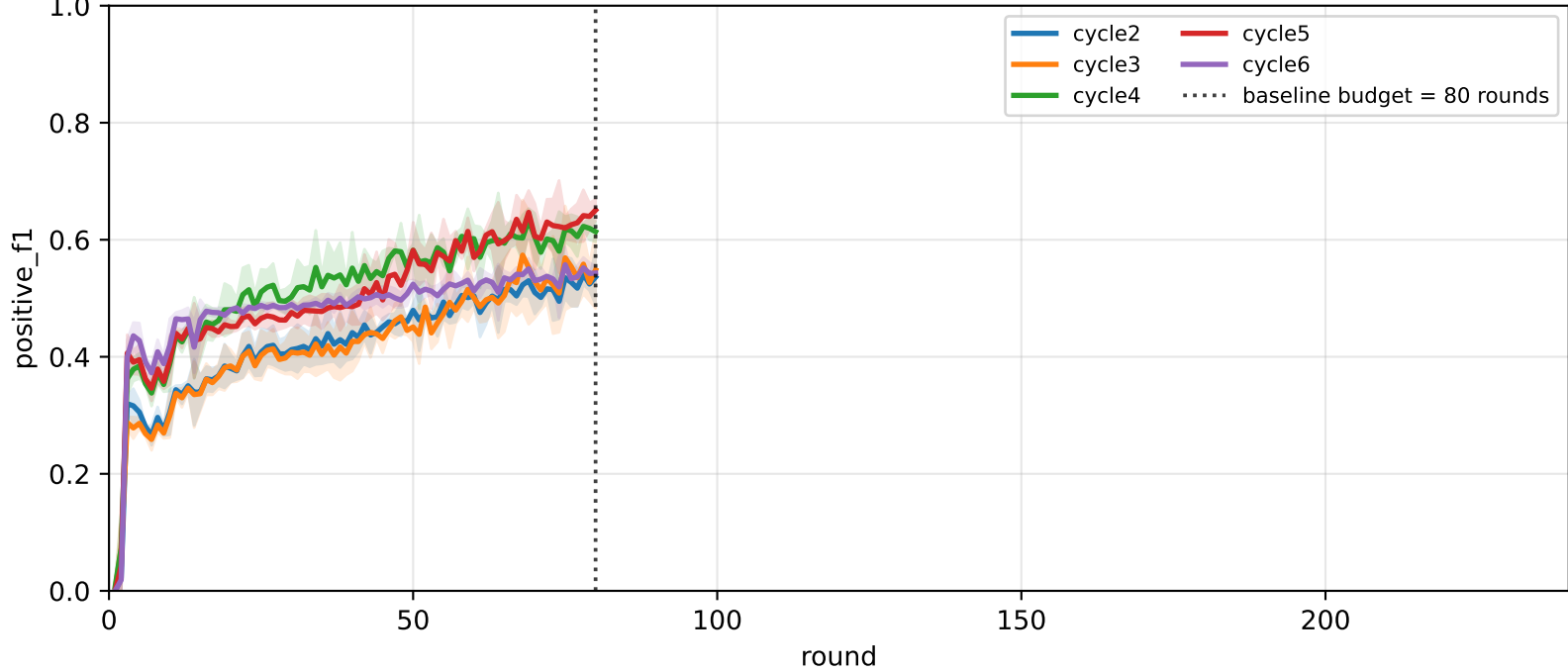
Validation loss comparison (all methods 80 rounds; APPLE dr_lr=0.001, mu=0.01, Lfrac=0.3)



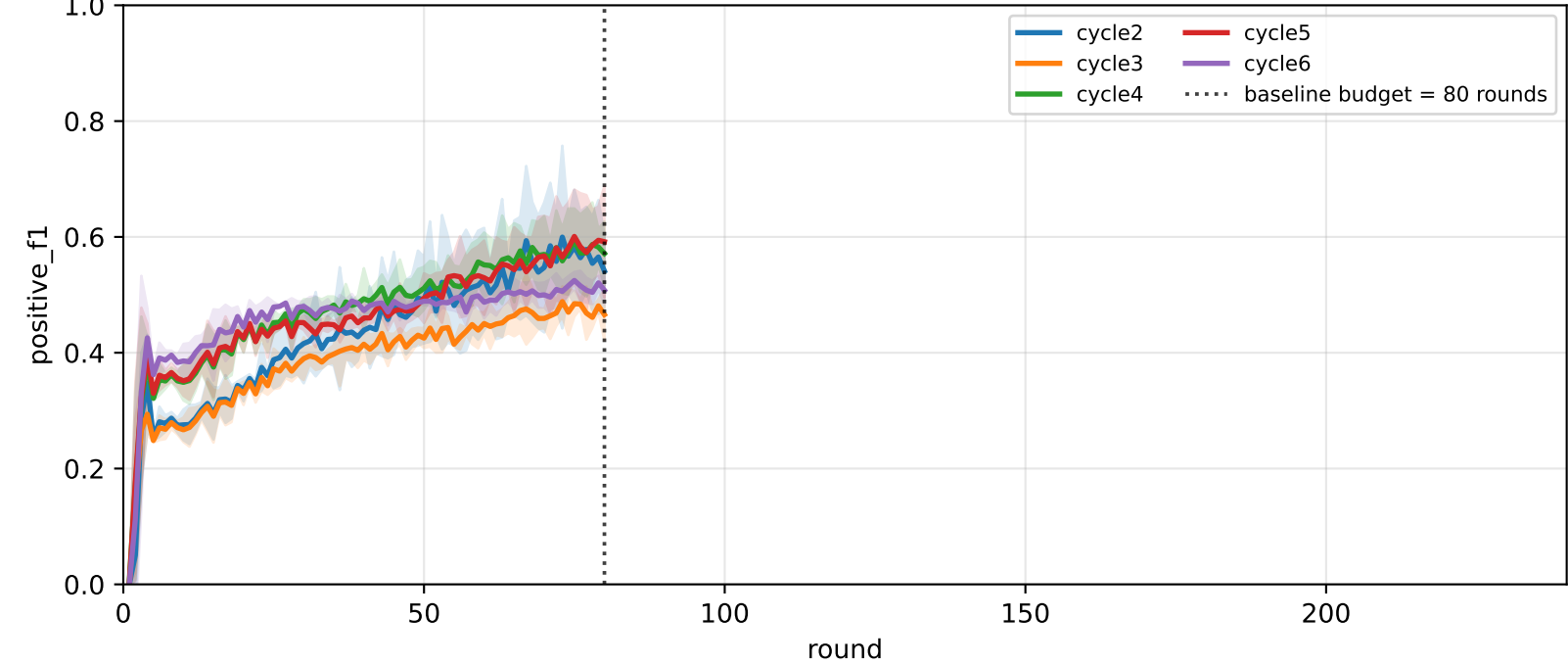
Pooled local centralized positive F1



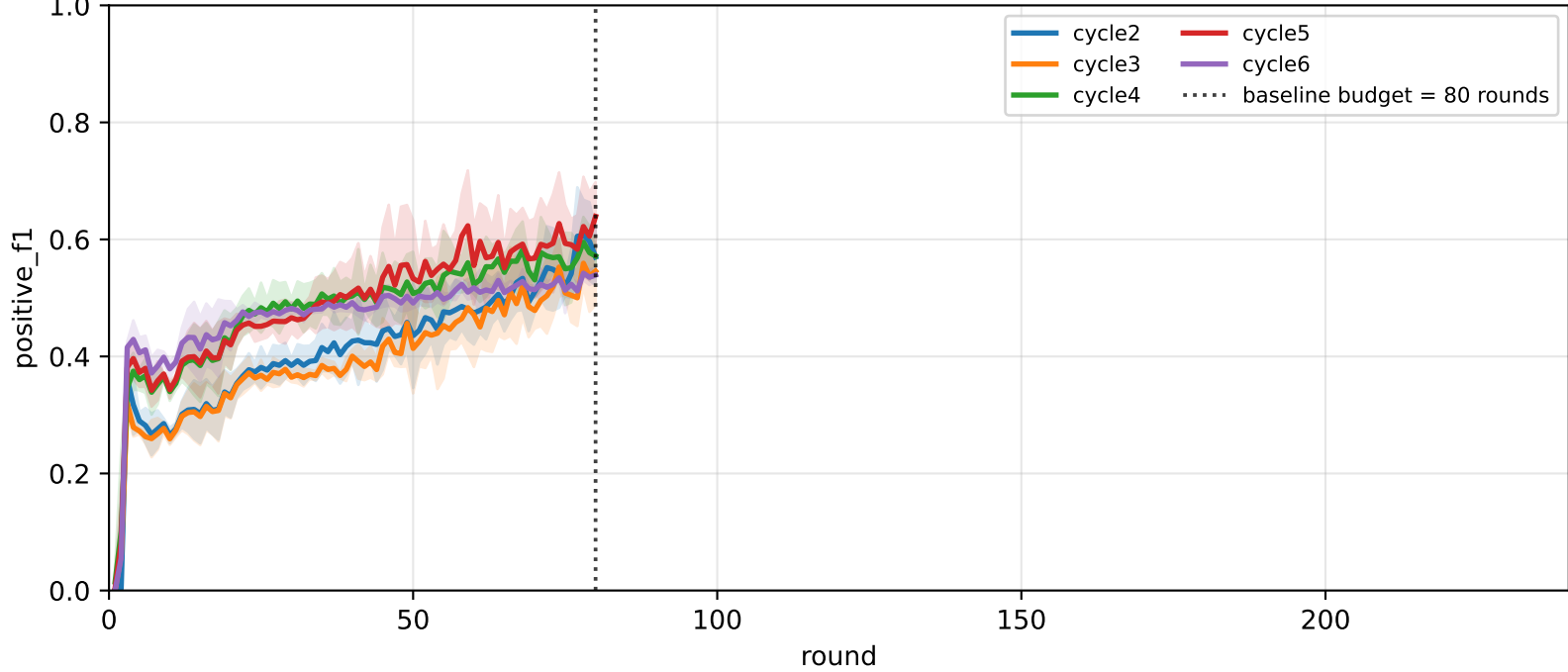
Pooled FedAvg positive F1



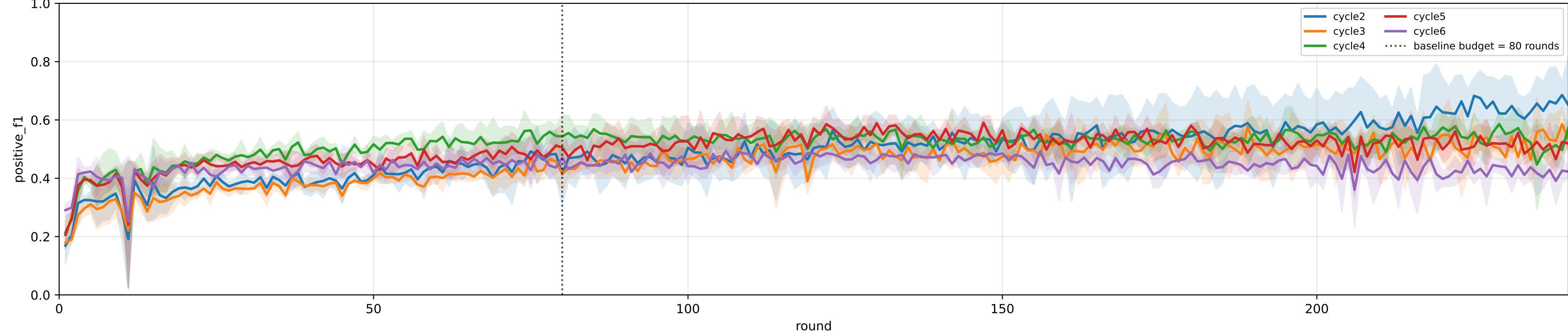
Pooled FedProx positive F1



Pooled GCFL+ positive F1



Pooled APPLE positive F1 (after-local selected model, equal 240-round budget)



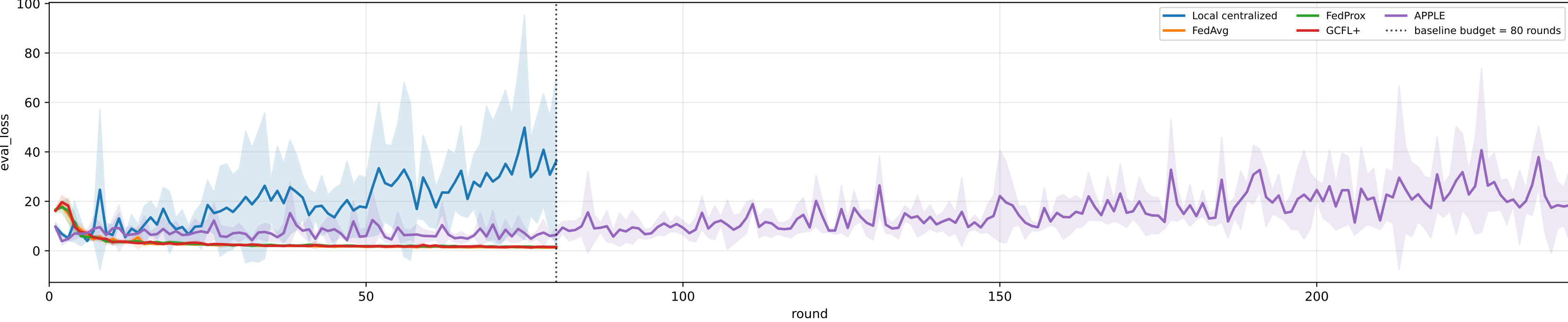
Controlled q-label heterogeneity: High | family aggregated / pooled performance |
family=q_task_cycle4 | n_clients=1 |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=q_task_cycle4 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 5.3%±0.0% / true 6.6% |
cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
heterogeneity=high | gamma=0.80 | task_jsd_mean=0.399 | task_jsd_median=0.399 | task_jsd_max=0.403 | family_centroid_jsd_mean=0.399

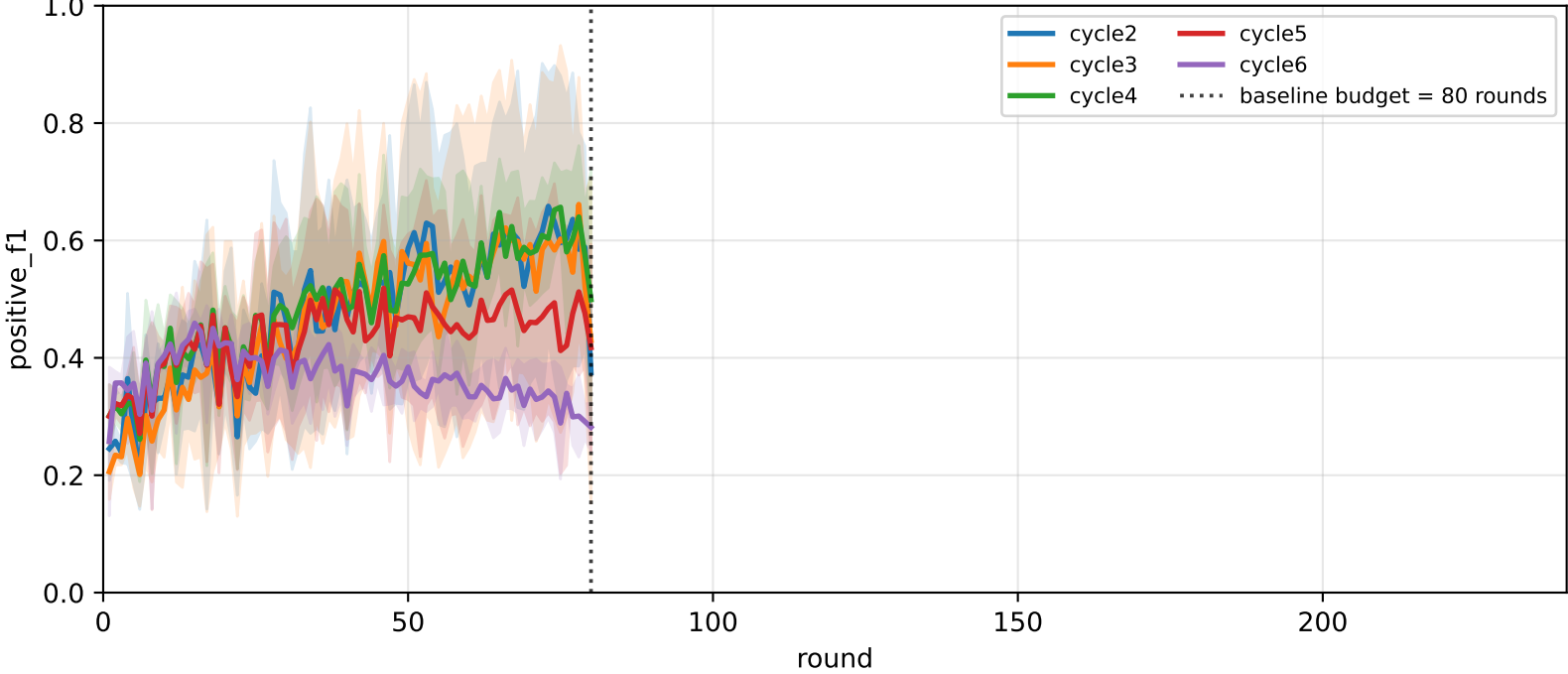
Best logged test performance (all methods 80 rounds; APPLE dr_lr=0.001, mu=0.01, Lfrac=0.3)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	20.9% ± 2.3%	21.0% ± 2.5%	16.7% ± 1.9%	21.0% ± 1.6%	20.9% ± 2.8%	20.3% ± 1.5%	26.1% ± 4.9%
FedAvg	54.7% ± 5.4%	55.2% ± 5.5%	50.3% ± 5.6%	60.7% ± 7.2%	57.1% ± 6.4%	58.0% ± 7.1%	49.8% ± 4.9%
FedProx	53.0% ± 2.8%	53.4% ± 2.8%	53.9% ± 3.0%	55.3% ± 3.6%	55.0% ± 5.8%	53.7% ± 6.0%	49.4% ± 0.9%
GCFL+	53.9% ± 1.2%	54.5% ± 1.3%	53.5% ± 4.6%	58.4% ± 4.6%	53.6% ± 4.7%	57.2% ± 5.5%	49.8% ± 1.0%
APPLE	31.1% ± 8.8%	31.0% ± 8.7%	26.5% ± 8.4%	31.4% ± 9.5%	31.9% ± 9.2%	31.1% ± 8.6%	34.4% ± 8.1%

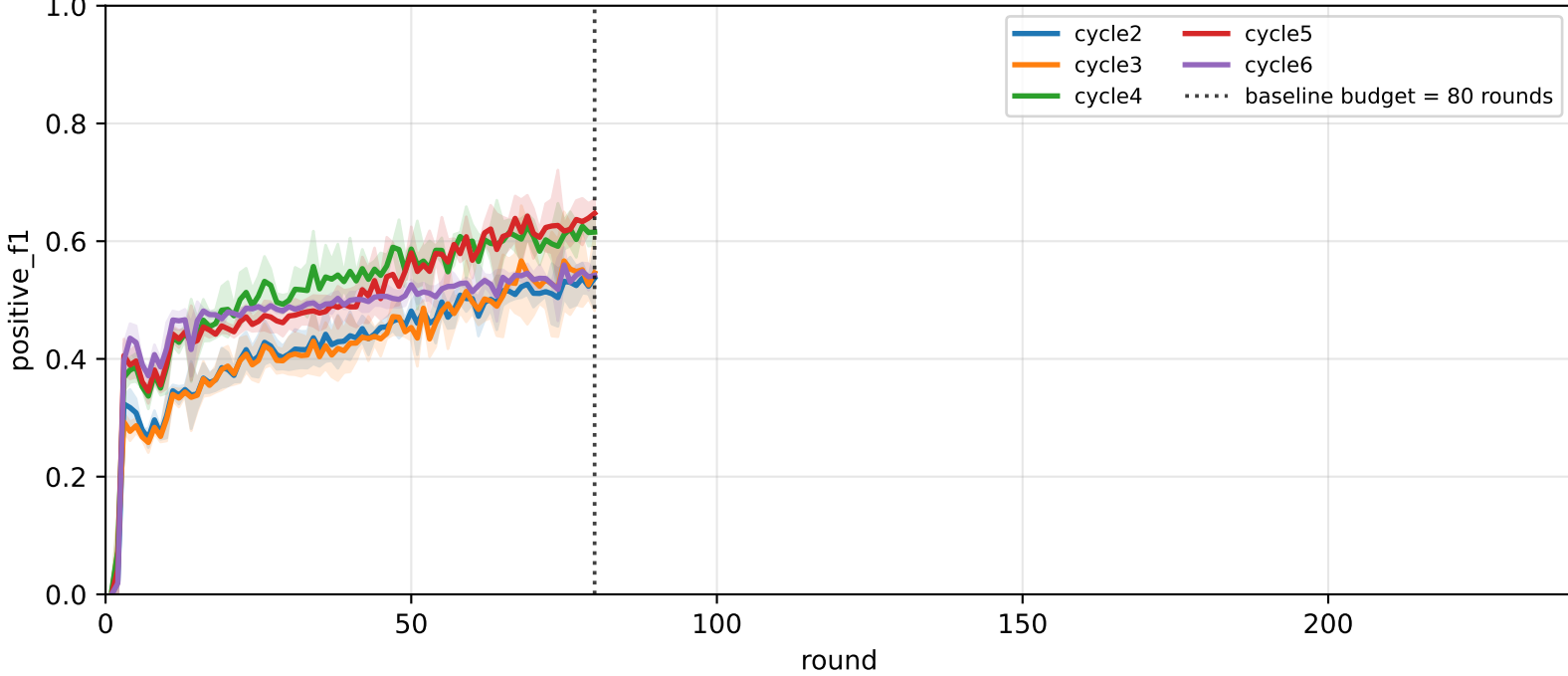
Validation loss comparison (all methods 80 rounds; APPLE dr_lr=0.001, mu=0.01, Lfrac=0.3)



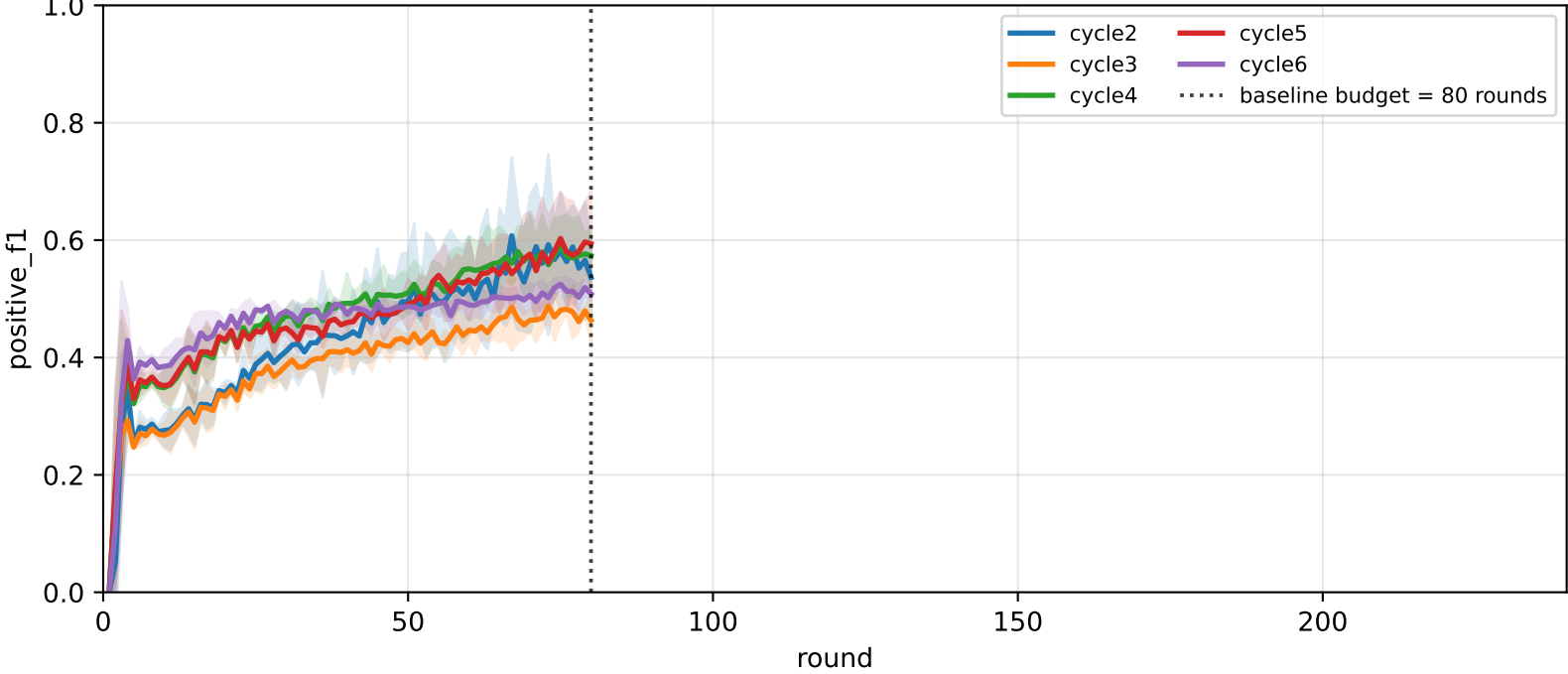
Pooled local centralized positive F1



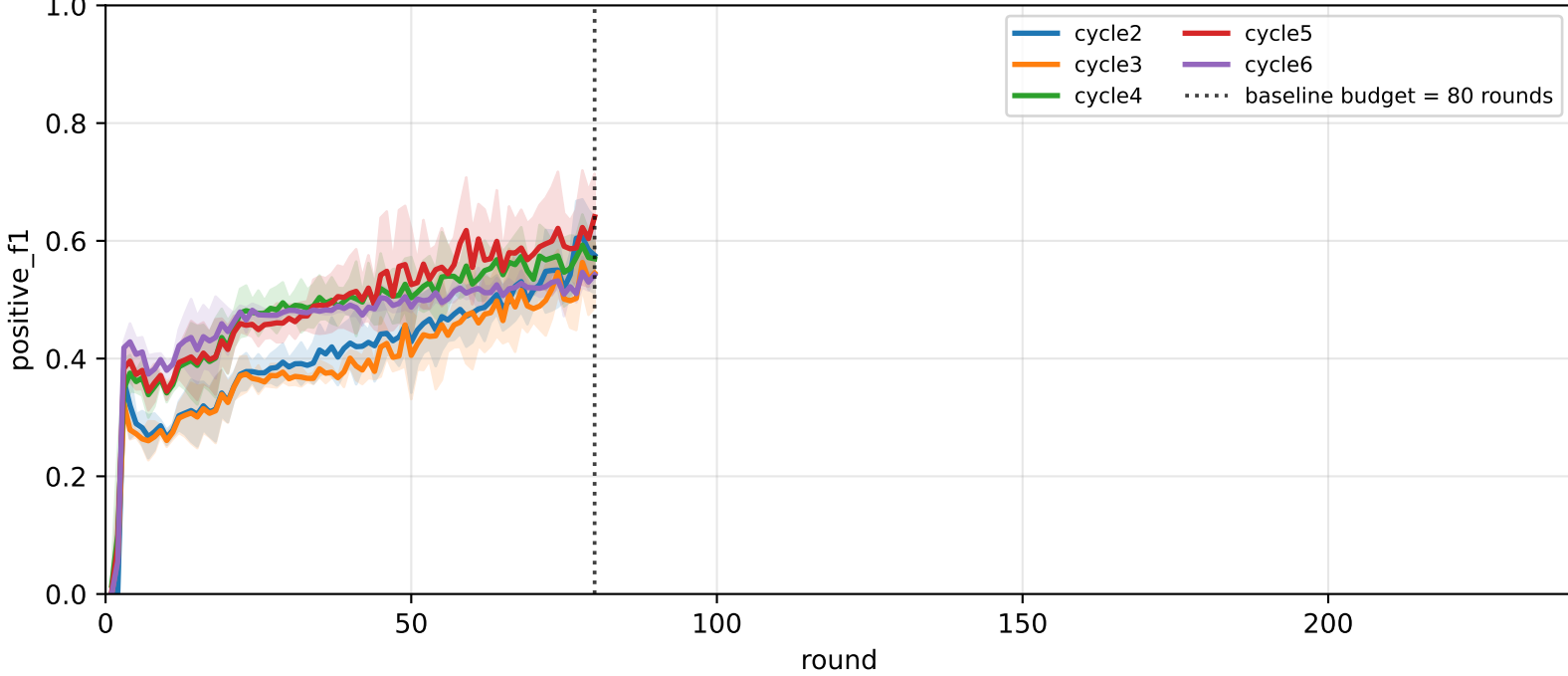
Pooled FedAvg positive F1



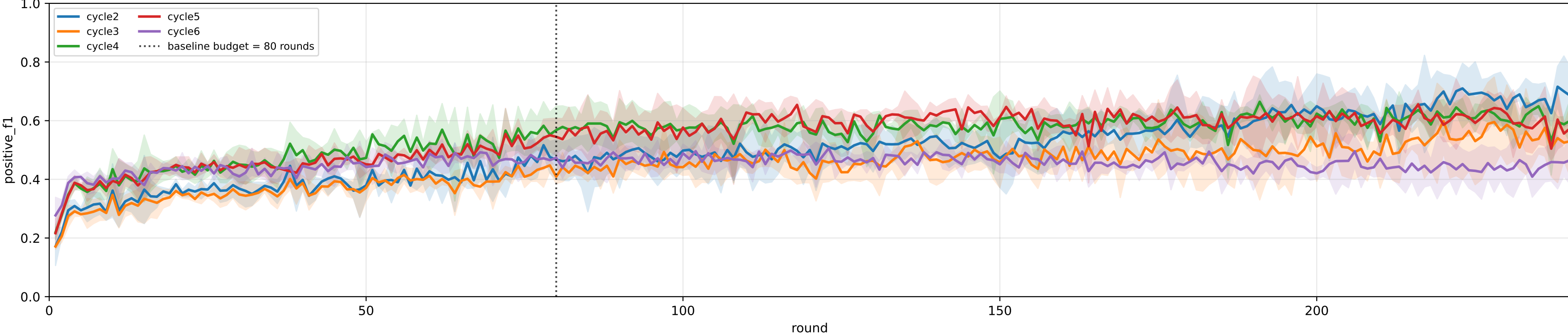
Pooled FedProx positive F1



Pooled GCFL+ positive F1



Pooled APPLE positive F1 (after-local selected model, equal 240-round budget)



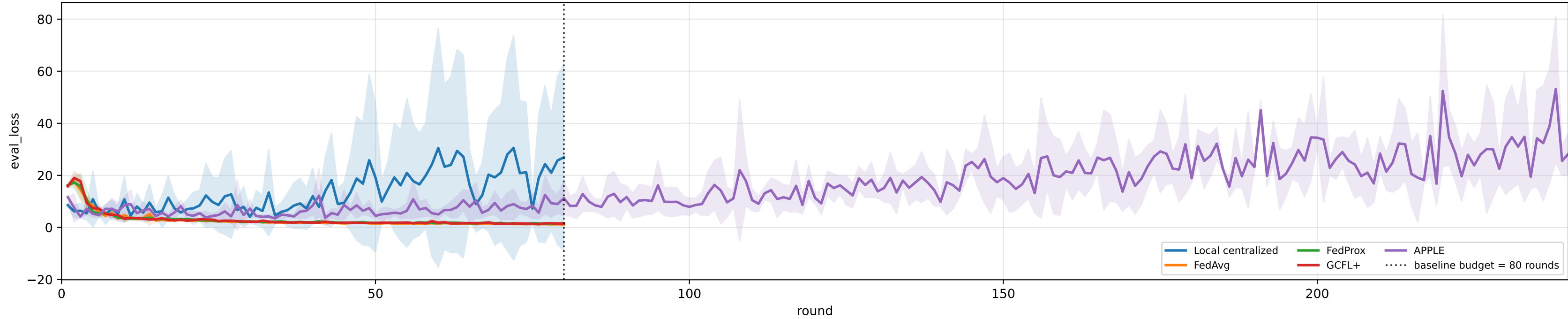
Controlled q-label heterogeneity: High | family aggregated / pooled performance |
family=q_task_cycle5 | n_clients=1 |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=q_task_cycle5 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% |
cycle5=visible 5.1%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3% |
heterogeneity=high | gamma=0.80 | task_jsd_mean=0.399 | task_jsd_median=0.399 | task_jsd_max=0.403 | family_centroid_jsd_mean=0.399

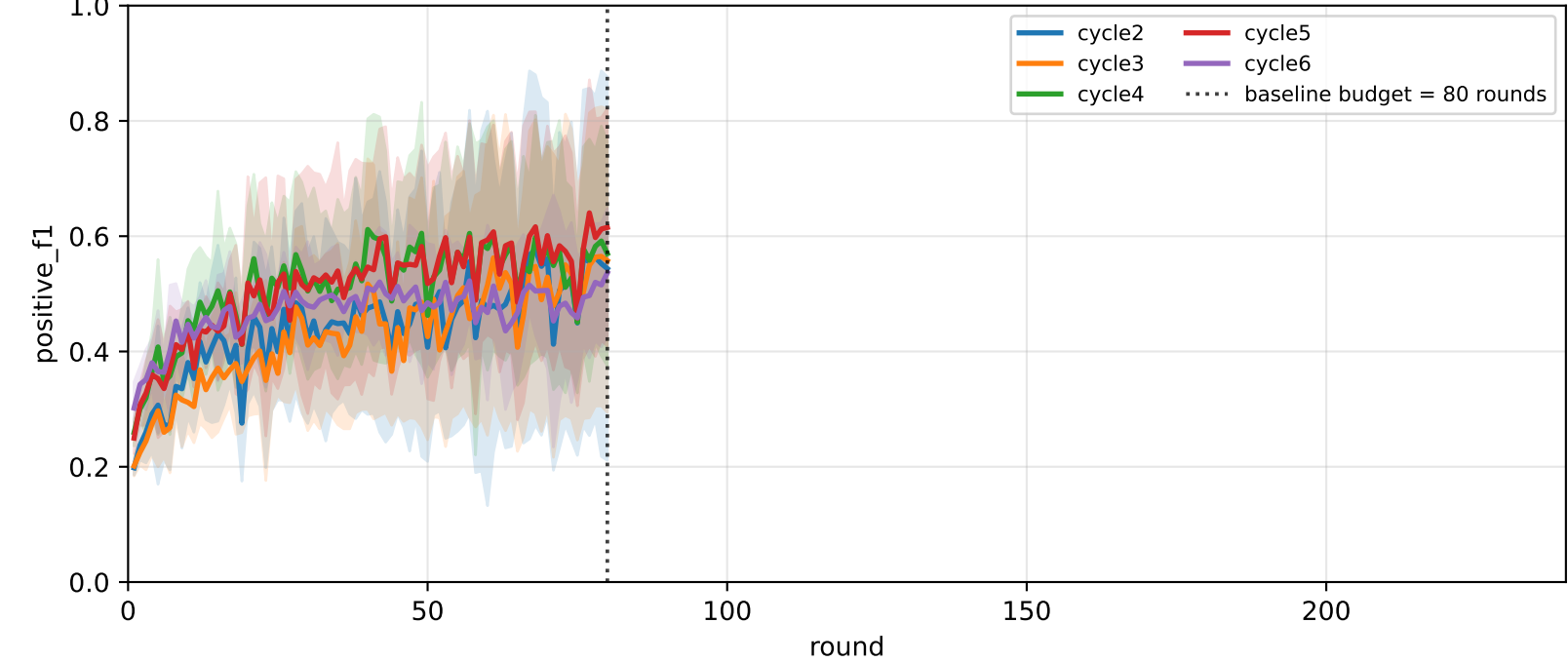
Best logged test performance (all methods 80 rounds; APPLE dr_lr=0.001, mu=0.01, Lfrac=0.3)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	34.0% ± 13.0%	34.3% ± 13.3%	29.8% ± 14.7%	32.8% ± 11.9%	34.7% ± 10.6%	37.8% ± 19.0%	36.2% ± 10.7%
FedAvg	54.2% ± 5.0%	54.7% ± 5.1%	50.1% ± 5.1%	60.4% ± 7.4%	56.9% ± 5.7%	56.6% ± 5.7%	49.4% ± 4.5%
FedProx	52.5% ± 2.7%	52.9% ± 2.7%	53.1% ± 3.0%	55.0% ± 3.4%	54.5% ± 5.6%	53.2% ± 5.8%	48.7% ± 1.0%
GCFL+	53.9% ± 1.4%	54.5% ± 1.6%	53.6% ± 4.8%	58.0% ± 4.2%	53.8% ± 4.2%	56.9% ± 5.7%	49.8% ± 1.0%
APPLE	30.5% ± 10.1%	30.4% ± 10.0%	25.8% ± 9.7%	31.3% ± 10.2%	31.5% ± 10.6%	30.0% ± 10.1%	33.5% ± 9.3%

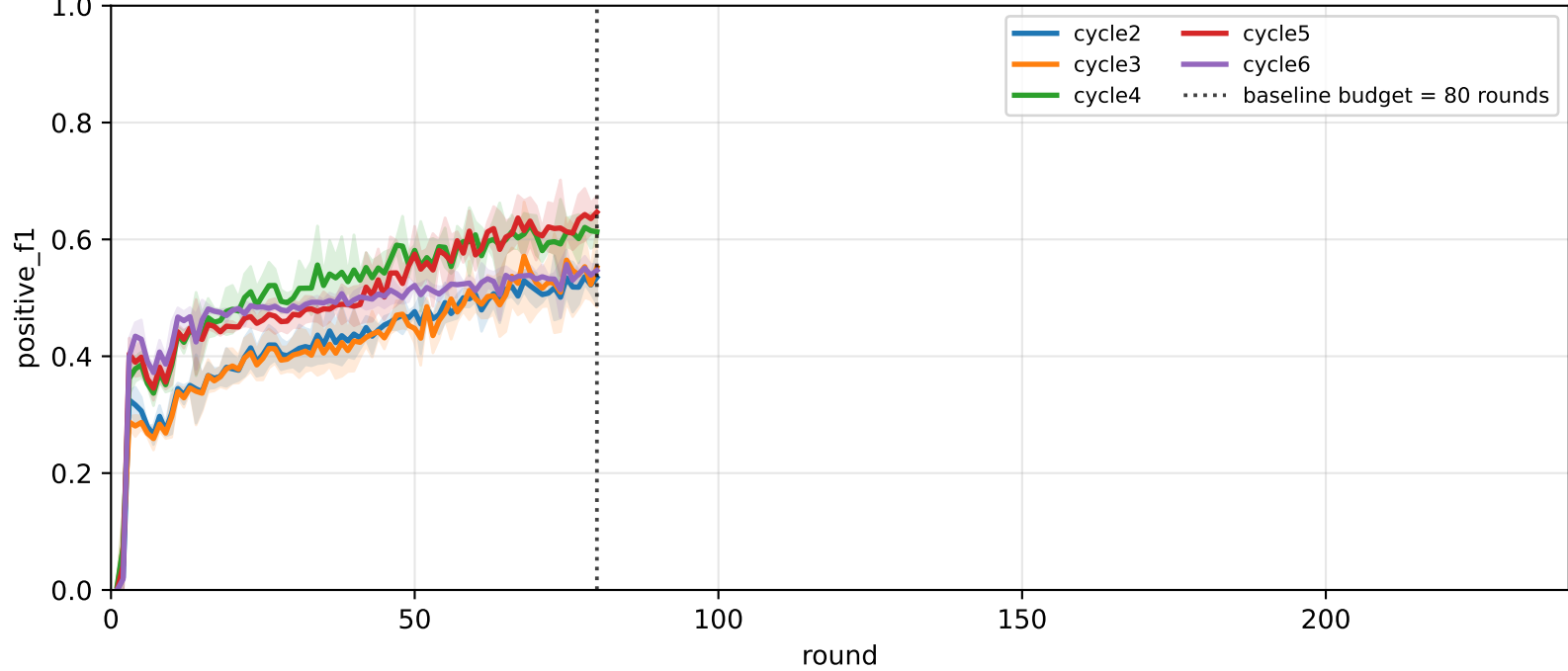
Validation loss comparison (all methods 80 rounds; APPLE dr_lr=0.001, mu=0.01, Lfrac=0.3)



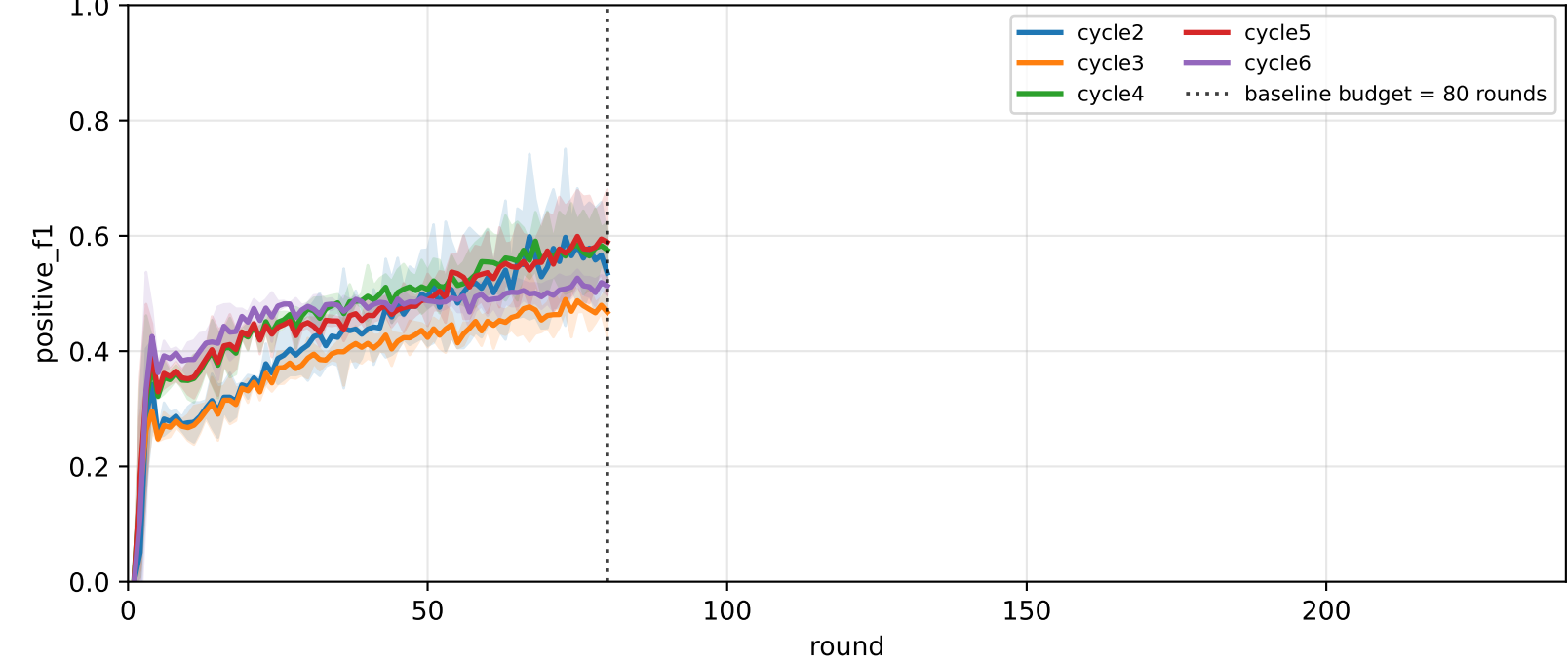
Pooled local centralized positive F1



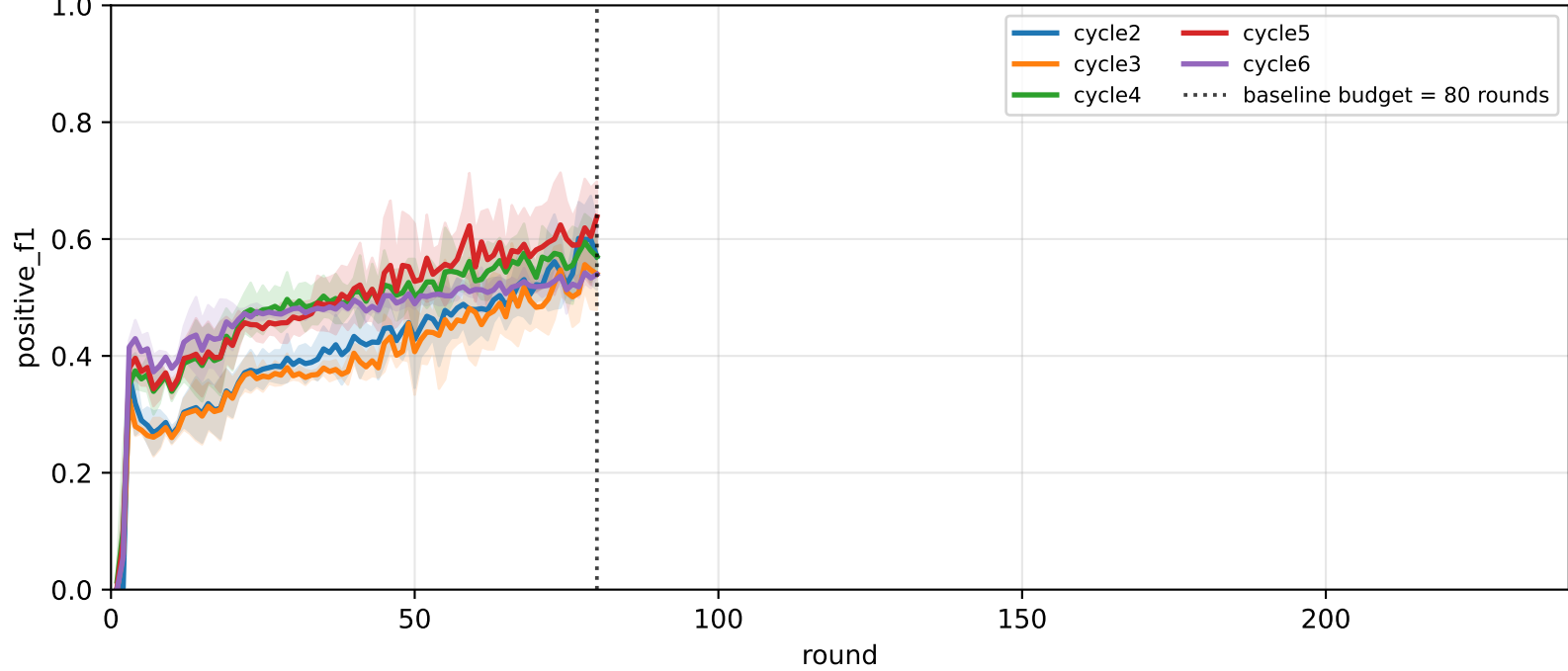
Pooled FedAvg positive F1



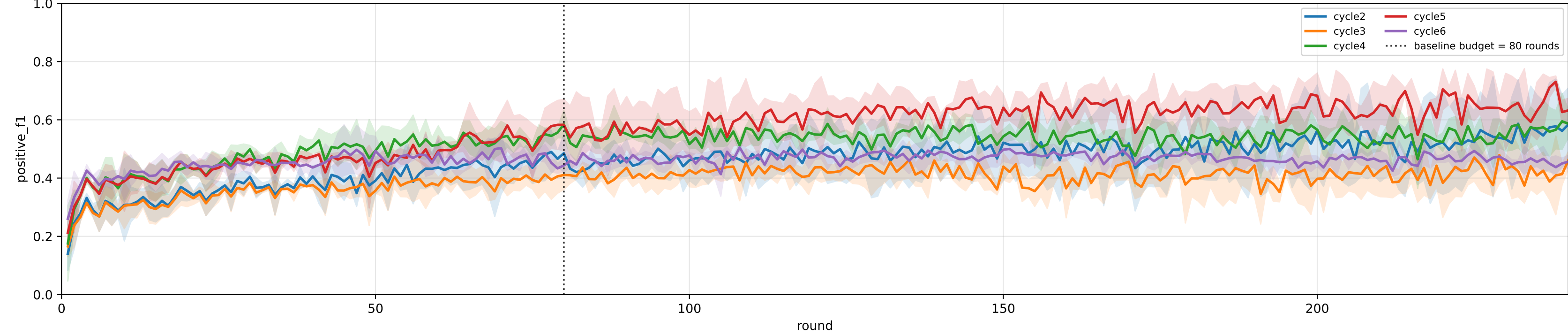
Pooled FedProx positive F1



Pooled GCFL+ positive F1



Pooled APPLE positive F1 (after-local selected model, equal 240-round budget)



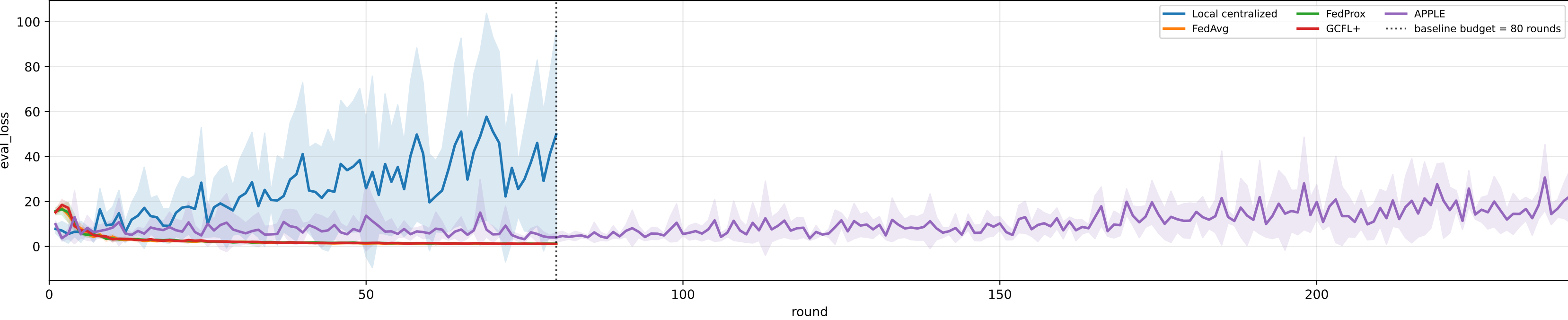
Controlled q-label heterogeneity: High | family aggregated / pooled performance |
family=q_task_cycle6 | n_clients=1 |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=q_task_cycle6 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% |
cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 5.0%±0.0% / true 6.3% |
heterogeneity=high | gamma=0.80 | task_jsd_mean=0.399 | task_jsd_median=0.399 | task_jsd_max=0.403 | family_centroid_jsd_mean=0.399

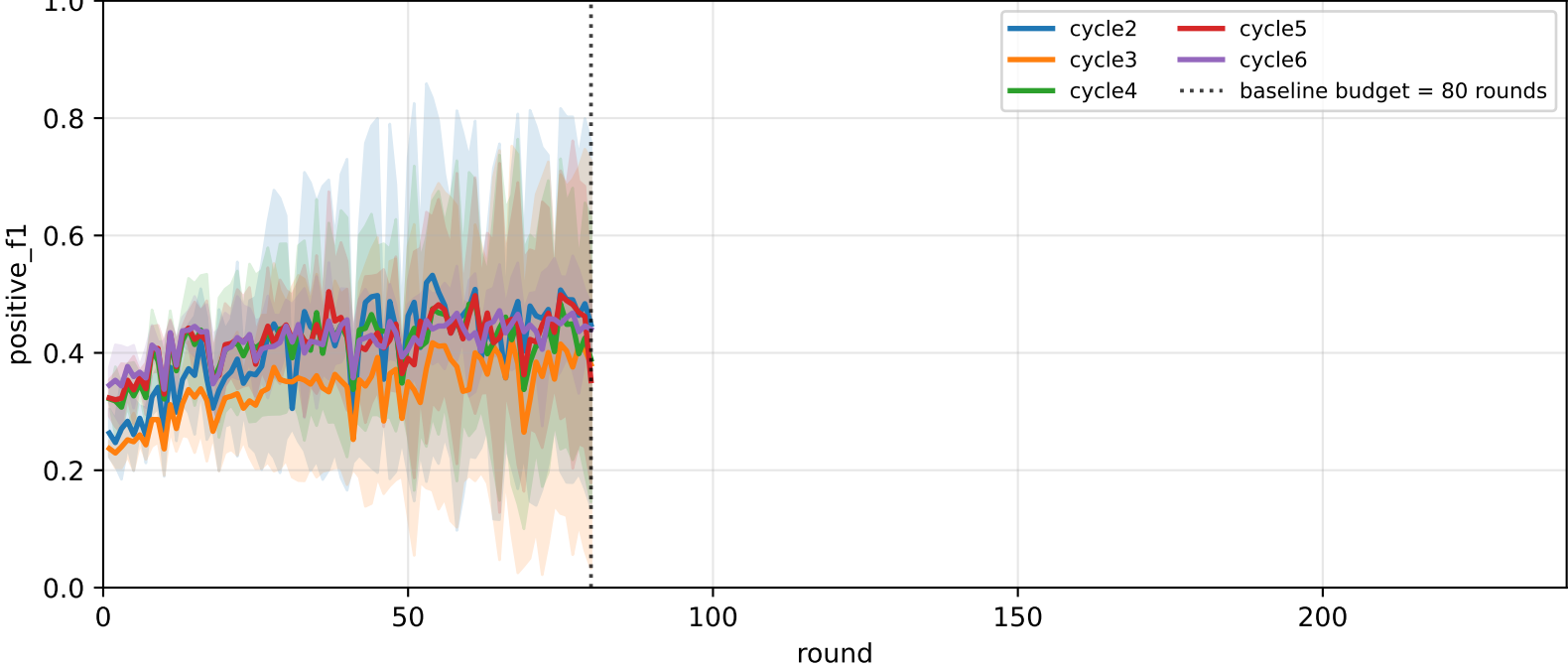
Best logged test performance (all methods 80 rounds; APPLE dr_lr=0.001, mu=0.01, Lfrac=0.3)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	33.5% ± 12.2%	34.3% ± 13.3%	36.4% ± 23.0%	36.9% ± 15.6%	32.4% ± 10.8%	32.2% ± 9.6%	33.5% ± 7.5%
FedAvg	54.6% ± 4.8%	55.1% ± 4.9%	50.3% ± 4.8%	60.8% ± 7.1%	56.7% ± 5.2%	58.0% ± 6.0%	49.6% ± 4.4%
FedProx	52.7% ± 3.3%	53.1% ± 3.3%	53.5% ± 2.4%	55.1% ± 4.0%	54.5% ± 6.3%	53.6% ± 6.4%	48.9% ± 1.5%
GCFL+	53.9% ± 0.7%	54.5% ± 0.8%	53.4% ± 4.5%	58.0% ± 3.2%	53.9% ± 4.6%	57.0% ± 5.3%	49.9% ± 1.1%
APPLE	31.6% ± 9.1%	31.5% ± 9.1%	26.5% ± 8.4%	31.9% ± 9.5%	33.0% ± 9.4%	31.6% ± 9.5%	34.7% ± 8.6%

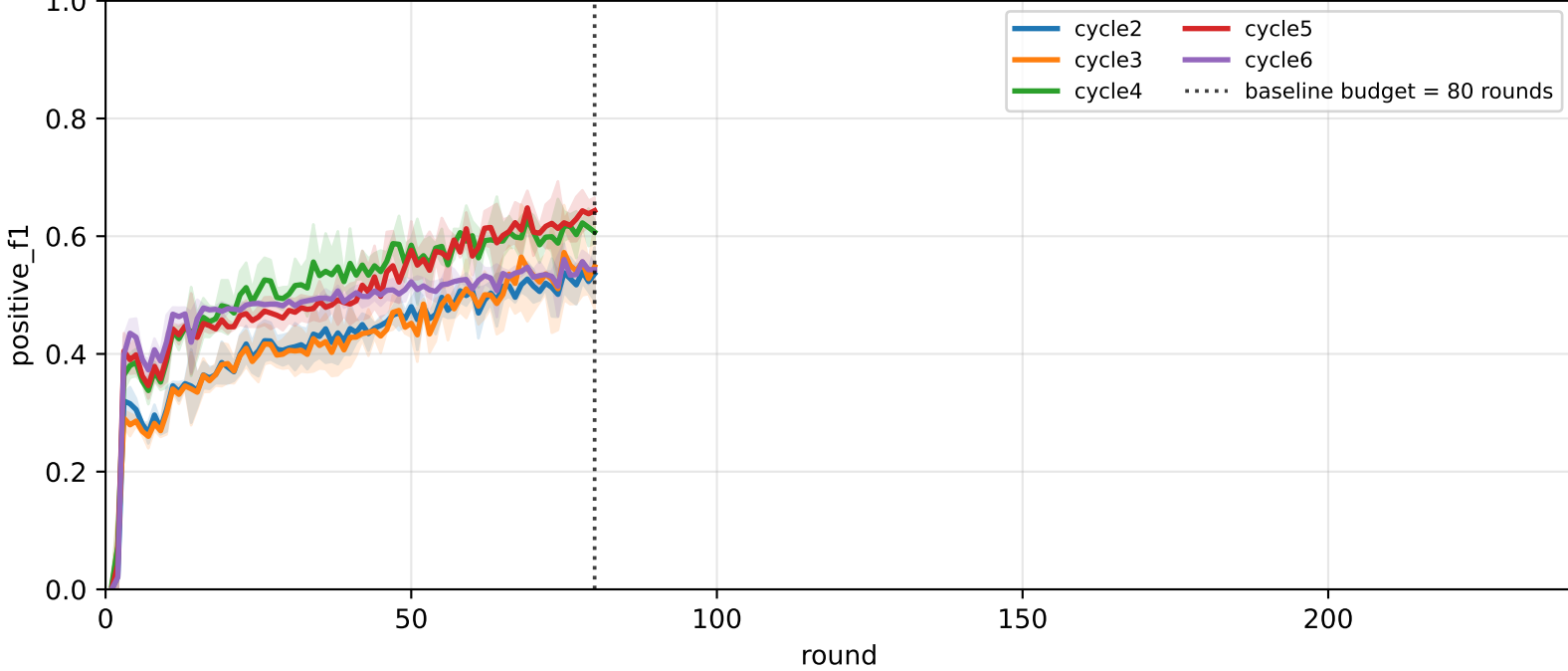
Validation loss comparison (all methods 80 rounds; APPLE dr_lr=0.001, mu=0.01, Lfrac=0.3)



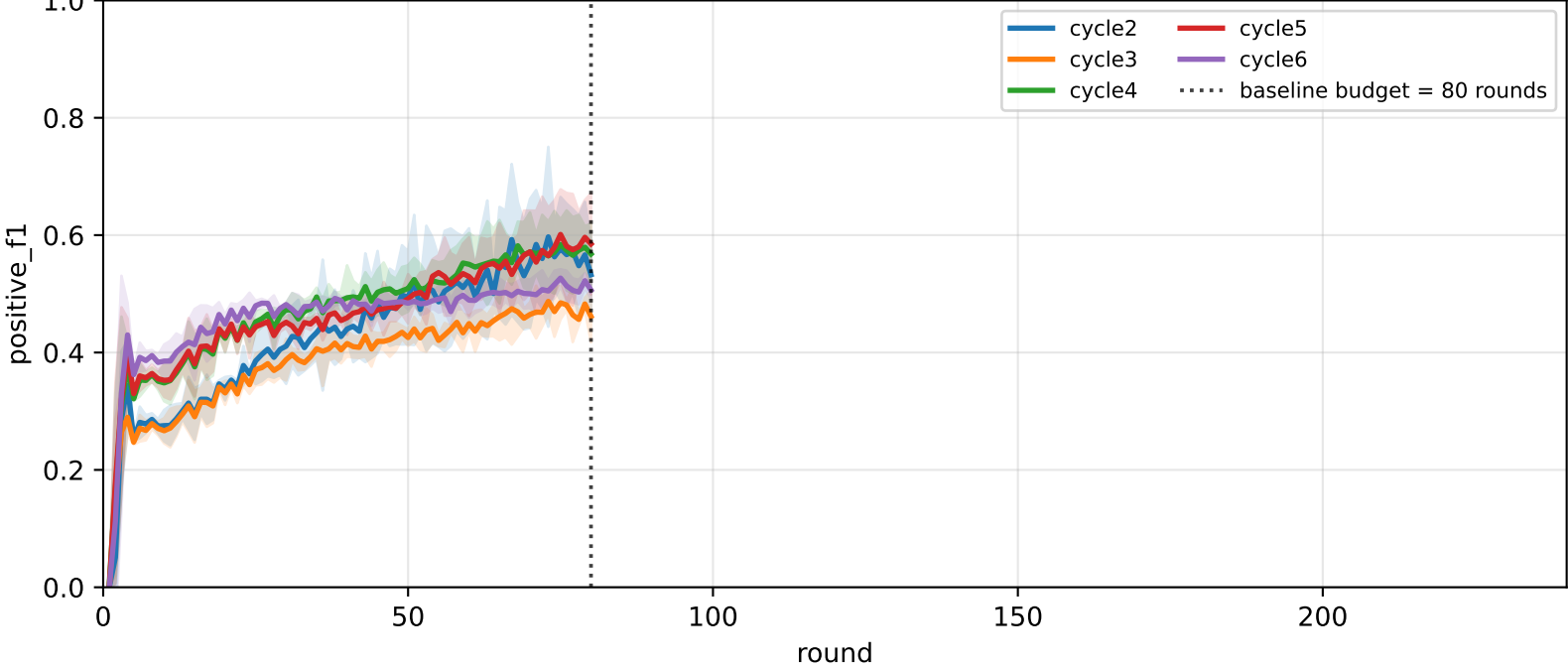
Pooled local centralized positive F1



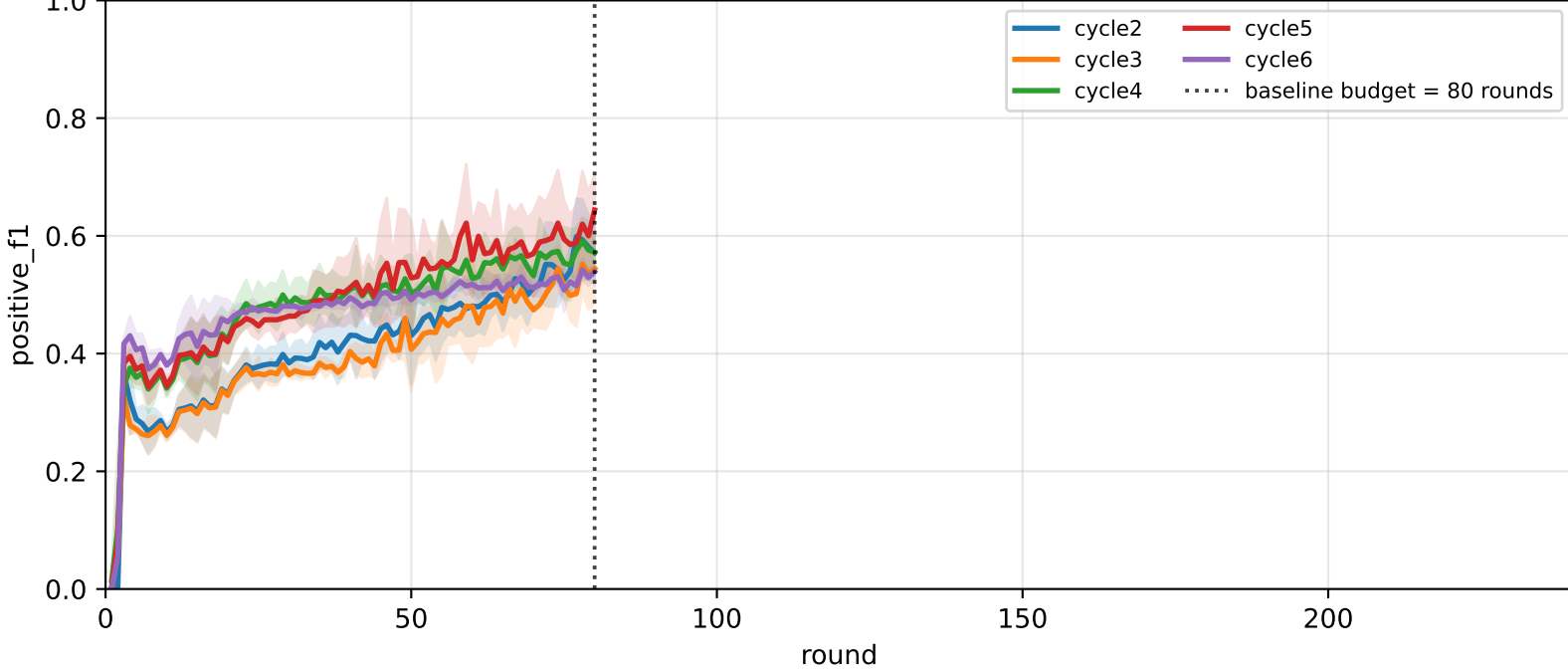
Pooled FedAvg positive F1



Pooled FedProx positive F1



Pooled GCFL+ positive F1



Pooled APPLE positive F1 (after-local selected model, equal 240-round budget)

